1. Module-Level Design

The system is divided into the following modules:

1. Document Processing Module:
Loads the PDF using PyPDFLoader.

2. Chunking Module:
Splits text into smaller chunks using RecursiveCharacterTextSplitter.

3. Embedding Module:
Converts text chunks into vector embeddings using MiniLM model.

4. Vector Storage Module:
Stores embeddings in ChromaDB.

5. Retrieval Module:
Fetches relevant chunks based on user query.

6. Query Processing Module:
Combines retrieved context and query to generate answer using LLM.

7. Graph Execution Module:
Controls flow using LangGraph nodes and edges.

8. HITL Module:
Handles escalation to human when confidence is low.

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2. Data Structures

Document:
Raw text loaded from PDF.

Chunk:
Small piece of document text (size ~500 characters).

Embedding:
Numerical vector representation of text.

State Object:

```
{
  "query": string,
  "response": string,
  "escalate": boolean
}
```

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3. Workflow Design (LangGraph)

Nodes:
1. Process Node:
   - Retrieves documents
   - Generates answer
   - Decides escalation

2. Output Node:
   - Displays chatbot response

3. Human Node:
   - Takes manual input from human agent

Edges:
Process → Output
Process → Human (if escalation)
Human → Output

State Flow:
Query → Process → (Decision) → Output / Human

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4. Conditional Routing Logic

If retrieved context is insufficient OR answer is unclear:
→ escalate = True

Else:
→ escalate = False

Routing:
If escalate = True → Human Node
Else → Output Node

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5. HITL Design

Escalation is triggered when:
- No relevant documents found
- Context is too small
- Model confidence is low

After escalation:
- Human agent provides response
- Response is returned to user

Benefit:
Improves reliability for complex queries

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6. API / Interface Design

Input:
```
{
  "query": "User question"
}
```

Output:
```
{
  "response": "Answer text",
  "escalate": true/false
}
```

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7. Error Handling

- If no documents found → return fallback + escalate
- If LLM fails → return context snippet
- If input is empty → return error message